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**How Much Agreement Survives Chance Correction?
LLM-as-a-Judge Reliability on Difficult Faithfulness Cases,
Measured Across Three Deployment Tiers**

Term Paper

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Abstract

LLM-as-a-judge is validated predominantly by raw agreement with human experts in the work reviewed here, a statistic that ignores the agreement two raters reach by coincidence. This study asks what survives when that coincidence is removed, and whether the remainder is enough to run a production quality gate. Three judges spanning three deployment tiers, a paid frontier API, a hosted open-weight model and an open-weight model running on a laptop, produced 7,012 judgments on FaithBench, a corpus of summaries selected because automatic detectors disagreed on them. The rubric, the analysis code and fourteen numbered predictions were each frozen before the test half of the corpus was read.

Raw agreement of 46.6 to 51.4 percent corresponds to a Cohen's kappa of 0.129 to 0.162, a difference of roughly 34 percentage points on every tier, consistent in direction and size with the difference reported on a different benchmark. All three judges agree with the reference labels less often than the corpus majority rule, a diagnostic baseline with zero specificity rather than a candidate gate. Paired bootstrap comparisons detected no difference in chance-corrected agreement between the three deployments, and the intervals were not narrow enough to rule out a difference of practical size. The errors are asymmetric: the judges reject faithful summaries rarely and miss roughly seven tenths of the unfaithful ones. Under a batch-level release policy, every tier therefore recommends shipping across a 40.6 point interval of thresholds over which the reference labels recommend blocking, and moving the threshold inside that interval does not align the two decisions.

Keywords: LLM-as-a-judge, faithfulness, hallucination detection, Cohen's kappa, chance-corrected agreement, evaluation methodology, pre-analysis plan

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1 Introduction and Research Question

1.1 Automated evaluation and the claim that legitimised it

A product built on a large language model can generate more text in a day than its team could read in a year. Since every generated passage carries some risk of stating something its source does not support, and since no organisation can review each one by hand, the reviewing is increasingly delegated to a second language model: one model writes, another grades. The arrangement is known as LLM-as-a-judge, and it now sits inside automated release pipelines, where its verdict decides whether generated text reaches a user.

The practice rests on an empirical claim. Zheng et al. (2023) reported that strong language models agree with human preference judgments more than 80 percent of the time, and observed that this is approximately the rate at which human annotators agree with one another. If a judge matches human raters about as well as human raters match each other, the reasoning goes, it can stand in for them. That comparison became an influential argument for using LLM judges as proxies for human evaluation, and it is invoked as a justification in much of the work reviewed in chapter 2.

The claim entered a field that had validated automatic evaluators the same way for years. Since Maynez et al. (2020) established hallucination as the central failure of abstractive summarisation, the model-based line from FactCC (Kryscinski et al., 2020) through SummaC (Laban et al., 2022) to AlignScore (Zha et al., 2023) and MiniCheck (Tang et al., 2024) has been validated by demonstrating that each detector's scores agree with human labels. Because these studies use agreement with human labels as validation evidence, the choice of agreement statistic materially affects their conclusions, and the studies named above report uncorrected agreement or correlation.

1.2 Why the validating statistic is contested

Raw agreement is the proportion of cases in which two raters gave the same answer, and its weakness is that raters also agree by coincidence. The effect is easiest to see in an extreme case. If nine summaries in ten are unfaithful, a judge that marks every summary unfaithful without reading a word matches the human labels 90 percent of the time. Raw agreement rewards that judge; the chance-corrected statistic introduced by Cohen (1960), standard in medicine and the social sciences ever since, scores it at zero, because its verdicts carry no

information about the labels. Real judges sit between the two, and the difference between what raw agreement reports and what a chance-corrected statistic reports is the quantity this paper measures.

The correction is not a simple improvement. Cohen's kappa is itself depressed when one class dominates, a behaviour Feinstein and Cicchetti (1990) named the first paradox of kappa and which Byrt et al. (1993) addressed with a prevalence-adjusted variant. A study reporting kappa on skewed data therefore has to show that its conclusion does not rest on that artifact, which is why both statistics are reported here.

Applied to LLM judges the correction appears occasionally, and at scale only once, in a preprint. Norman, Rivera and Hughes (2026) find that it removes 33 to 41 points of apparent agreement on MT-Bench, and that model rankings move by as much as fourteen positions once it is applied. Two further results suggest that judges are less interchangeable than the practice assumes: Nasser (2026) reports inter-judge agreement close to zero across nine judges, with a Krippendorff's alpha of 0.042, while a classifier recovers which judge produced a set of scores 77.1 percent of the time from the scores alone; and Yang et al. (2026) find that improvements to a judge do not reliably transfer across tasks.

1.3 What remains open

Three gaps follow. The first is the corpus: the difference Norman et al. report was measured on general preference judgments, where the candidate answers are often plainly different, and whether it holds where detection is genuinely hard is unestablished.

There is reason to expect a difference. Thakur et al. (2025) apply the same correction to thirteen judges marking TriviaQA answers and report kappa of 0.79 and 0.84 for their two strongest, which is excellent agreement rather than deflated agreement, against an inter-human kappa of 0.96 on the same items. On FaithBench, a corpus assembled by Bao et al. (2025) from summaries on which existing detectors disagreed, the best detector they tested reaches 62.31 percent balanced accuracy against a 50 percent random baseline. Whether chance correction is destructive may therefore depend on how hard the underlying task is, and that dependence has not been measured directly.

The second is the variable compared. Published comparisons vary the judge model, whereas the decision an organisation faces is a deployment decision: whether to send data to a

paid proprietary API, to a hosted open-weight model, or to a model running on hardware it already owns. Those options differ in cost, latency and data exposure as well as capability, and whether they differ in judgment quality is a question the model-centred comparisons do not address.

The third concerns what agreement is used for. An agreement statistic summarises how often judge and human coincide across a corpus, while a release gate is worth what it is worth on the failures it catches. A judge can score acceptably on the first property and detect too little to support the second.

1.4 Research question and scope

This study addresses those gaps with one question:

On cases where automated faithfulness detection is genuinely difficult, how much of the agreement between LLM judges and human experts survives chance correction, and is the remaining judgment sufficient to support a production quality gate?

Two phrases carry the weight. Survives chance correction distinguishes the question from the validation practice described above, and production quality gate makes detection of the unfaithful cases the decisive quantity rather than overall accuracy, since a gate that never fires provides no protection. Four sub-questions follow, and they are referred to by these labels throughout.

SQ1 How far do raw agreement, Cohen's kappa and a prevalence-adjusted statistic diverge on identical judgments?

SQ2 How does each judge compare against the agreement the human annotators reach with one another?

SQ3 What proportion of the summaries the experts identified as unfaithful does each judge detect, and what proportion of faithful summaries does it wrongly reject?

SQ4 Under a batch-level release policy, does the ship-or-block decision agree with the decision the human reference labels imply, and does it depend on which tier the judge runs in?

SQ4 names a specific policy, and the whole of the gate result depends on it. A batch of summaries is judged, the share flagged unfaithful is computed, and the release is blocked

when that share exceeds a threshold. The same rule applied to the reference labels gives the comparison decision. Section 4.4 states the policy formally and section 5.5 states what it cannot decide. The scope is otherwise deliberately narrow: the study measures judges against fixed human labels in an existing corpus of English news summaries, and it does not generate summaries, train a model or propose a new detector.

2 Theory and State of Research

2.1 Faithfulness, and what it is not

Faithfulness is a relation between a summary and its source: does the source support every claim the summary makes? It is not truth. A summary can be faithful to a wrong source, and it can be unfaithful while stating a fact. The standard split is between intrinsic hallucination, where the summary contradicts the source, and extrinsic hallucination, where it adds information the source does not contain (Huang et al., 2023; Bang et al., 2025).

One terminological trap has to be closed at the outset. Siegel et al. (2025) use faithfulness for self-explanation faithfulness, meaning whether a model's stated reasons actually caused its answer, which is a different relation with the same name. This paper measures summary-to-source faithfulness throughout.

2.2 The measurement tradition before LLMs

Maynez et al. (2020) established hallucination as the central failure of abstractive summarisation and showed that entailment-based measures track human judgment better than ROUGE does. Their Table 10 reports that between 73.1 and 79.3 percent of the summaries produced by every system they tested contained at least one hallucinated word, counting a word as hallucinated only when all three annotators agreed on it. One row of that table returns in chapter 5: the human-written gold reference summaries were themselves annotated as hallucinated at 76.9 percent, indistinguishable from the systems they were being used to evaluate. The cause there is an artifact of how XSum was built, since its references are the article's own first sentence, but the general observation survives the specific cause. A benchmark whose reference labels encode something other than ground truth is documented and expected, not a peculiarity of one corpus.

The model-based line runs from there: FactCC (Kryscinski et al., 2020) as the first learned consistency checker, SummaC (Laban et al., 2022) as the strongest pre-LLM baseline built

on natural language inference, the question-answering family represented by QAGS (Wang et al., 2020), and AlignScore (Zha et al., 2023) and MiniCheck (Tang et al., 2024) as the current cheap baselines. Each of those five was validated by correlating with human judgment, so the validation statistic matters for the line of work they form, and in each case the headline statistic is an uncorrected correlation or agreement.

2.3 LLM-as-a-judge and the claim that legitimised it

Zheng et al. (2023) is the anchor. The claim that opened the practice is roughly 80 percent agreement between strong LLM judges and human experts, described as comparable to the agreement humans reach with each other. That number is raw agreement: the proportion of cases where judge and human said the same thing, uncorrected for coincidence.

A great deal has been built on top of it. Judge ensembles, rubric engineering, position-bias mitigation, self-consistency and reference-anchored judging all aim to make the judge better at its task, and the statistic commonly used to show that it is better has not changed with them. That is the hinge of this chapter, and the sections after it are the evidence.

The correction is not unknown in this literature. Thakur et al. (2025) report kappa beside percent agreement and argue for it explicitly. What has not followed is a change in convention, and the headline validating figure in the work reviewed here remains raw agreement.

2.4 Judges are not interchangeable instruments

Three recent studies arrive at the same place from different directions. Nasser (2026) reports 3,240 evaluations across nine judges and finds inter-judge agreement close to zero, with a Krippendorff's alpha of 0.042, while a classifier identifies which judge produced a set of scores 77.1 percent of the time from the rubric scores alone. Each judge is internally consistent while disagreeing with every other judge, because each applies its own durable notion of quality, and the consequence Nasser draws is worth stating plainly: averaging their scores produces a verdict that corresponds to no judge's actual values.

Yang, Hou and Yang (2026) hold the responses fixed and swap the judge across four datasets, and find that judge upgrades do not transfer reliably. Norman, Rivera and Hughes (2026) examine roughly 541,000 judgments and report a difference between exact-match agreement and Cohen's kappa of 33 to 41 percentage points, with rankings shifting by up

to fourteen positions. Chapter 4 measures the same difference on a different benchmark and a different task.

2.5 Measurement theory from clinical epidemiology

Cohen (1960) defines kappa. Two raters labelling a binary property agree some of the time by coincidence alone, and how often depends on how skewed each rater's label distribution is. Kappa rescales the observed agreement against the agreement expected from those distributions, so it estimates agreement beyond that expectation rather than identifying which individual agreements were coincidental. Feinstein and Cicchetti (1990) describe the first of two paradoxes that follow, namely that a high observed agreement can be drastically lowered by a substantial imbalance in the table's marginal totals. That paradox applies directly to the corpus used here, which is why the prevalence-adjusted variant is reported beside kappa in chapter 4.

Byrt, Bishop and Carlin (1993) propose the prevalence-adjusted and bias-adjusted form the later literature named PABAK, and their own recommendation is what the analysis plan implements: when agreement studies are compared it can be misleading to report kappa alone, so quantitative indicators of bias and prevalence belong beside it. Zec et al. (2017) provide open-access worked examples of kappa collapsing under high unbalanced prevalence.

2.6 The dataset and its critics

FaithBench (Bao et al., 2025) is built from summaries on which automatic detectors disagreed, which is what makes it an instrument for isolating difficulty rather than a corpus that inherits it. It annotates at span level under four levels, and two of the definitions matter later. Questionable means the span is not clearly a hallucination and classification may differ depending on whom you ask. Benign means it is clearly a hallucination but is supported by world knowledge, common sense or logical reasoning, so a reader finds it acceptable. The authors do not nominate a single binarisation and report detectors under three different schemes, which is the direct justification for this paper reporting all three as well. Their best evaluated detector reached 62.31 percent balanced accuracy against a 50 percent baseline.

Ding, Luo, Cao and Luo (2025) argue that existing schemes leave unspecified how much external world knowledge an annotator may bring, and add an intermediate category for claims whose verification requires knowledge the source does not contain. Their Appendix A examines FaithBench directly, and the important detail is that the labels they dispute are unanimous ones, in both directions. The objection is therefore not that the grey labels are vague at the margin. It is that agreement among annotators does not establish correctness. Chapter 5 carries that consequence rather than rebutting it.

FaithJudge (Tamber et al., 2025) is the closest prior work, and any comparison of the numbers has to start from the design difference. FaithJudge supplies the judge with the other nine annotated summaries of the same source document, spans and labels included. With the judge model held constant, balanced accuracy on FaithBench rises from 68.8 percent under zero-shot prompting to 84.0 percent. Same model, two prompt designs, a gain of 15.2 points. Most of that gap is the scaffolding rather than the judge, and that configuration is unavailable in production by construction, because a gate meets each new document without nine annotated peers of it.

2.7 The gap, and what it demands of a method

Judge quality is actively researched and steadily improving, and section 2.3 listed the safeguards. Judge validation has not moved with it: in the work reviewed here those safeguards are still validated by the uncorrected statistic, and the one large study that applied the correction found it removes a third of the reported agreement.

The gap is therefore not that a statistic is missing, since kappa has existed since 1960. It is that the reviewed evidence does not consistently report the chance-corrected and class-specific measures a reader would need in order to judge whether a given judge is suitable for a release workflow. Nasser's nine mutually disagreeing but internally consistent judges, and Yang et al.'s upgrades that fail to transfer, are both unintelligible under a single uncorrected agreement number and unsurprising once the number is corrected. Two things follow for the method. Because the gap is a gap in measurement, the measures have to be derivable from the question rather than from whatever the tooling makes convenient. And because the finding is going to be uncomfortable, the study has to fix its own claims before it sees its own data.

3 Method

3.1 Deriving the measures

The measures are derived using the Goal Question Metric framework (Basili, Caldiera & Rombach, 1994). GQM justifies the choice of measures rather than narrating the order in which they were chosen: the rubric had already been frozen when this derivation was written down.

The goal, in Basili's template: analyse LLM-as-a-judge verdicts on summary faithfulness, for the purpose of evaluation, with respect to agreement with human expert labels after chance correction, from the viewpoint of a practitioner deciding whether to deploy an automated quality gate, in the context of abstractive summarisation of English news on cases selected for detector disagreement.

Four questions follow. Q1 asks how much observed agreement survives chance correction, answered by raw agreement (M1), Cohen's kappa (M2), the difference between them (M3) and PABAK (M4), the last because kappa alone is open to the Feinstein and Cicchetti objection under this corpus's roughly 69 to 31 imbalance. Q2 asks whether chance-corrected agreement differs across tiers and adds the variance of kappa across three repeats (M5), reported in section 4.1 as an instrument property, since a difference between tiers is interpretable only against the instability of one tier. Q3 asks how a judge fails, using sensitivity (M6) and specificity (M7) reported separately, the flag rate (M8) and detection by severity (M9). Q4 asks whether the remaining agreement suffices for a gate under the batch-level policy defined in section 1.4, using the width of the interval of thresholds over which the judge-based and reference-based decisions disagree (M10). Three further metrics establish instrument validity: parse-failure rate (M11), prompt constancy (M12) and version drift (M13).

The four sub-questions stated in chapter 1 are reader-facing, and the derivation above produces four metric-facing ones. Table 1 joins them, because a reader following the argument otherwise loses the thread at that joint.

Sub-question	GQM	Metrics	Reported in
SQ1 divergence between the three statistics	Q1	M1 to M4	4.2
SQ2 comparison against human agreement	Q1	M1 vs 82.0%	4.2
SQ3 catch rate and false-rejection rate	Q3	M6 to M9	4.3

Sub-question	GQM	Metrics	Reported in
SQ4 the batch-level release decision	Q4, Q2	M10, M2	4.4

Table 1: Crosswalk between the locked sub-questions, the GQM questions and the metrics that answer them. Every sub-question has a section of chapter 4 that answers it, and no metric exists that no question asked for.

3.2 The hypotheses, and when they were fixed

Fourteen numbered, quantified and falsifiable predictions were committed to version control before the test split was read, each with a threshold that would have counted as failure. Nothing in the list has been altered since; a verdict column was appended and nothing else touched, and the commit holding the pre-test state is cited in Appendix C so the claim can be checked by diff. The five hypotheses in Table 2 group those predictions and were written afterwards for readability, so the grouping is presentation rather than evidence.

	Hypothesis	GQM	Predictions
H1	Chance-corrected agreement is far below raw agreement, by an amount of the same order as the 33 to 41 points reported on a different benchmark	Q1	P1, P2, P5, P13
H2	Deployment tier does not predict chance-corrected agreement; the three arms are not distinguishable	Q2	P3, P4
H3	Failure is asymmetric: high specificity, low sensitivity, worst on the ambiguous severity band	Q3	P6, P8, P9, P14
H4	An interval of release thresholds exists over which every tier ships while the reference labels block	Q4	P10
H5	The analyst's choice of binarisation scheme moves the result as much as the choice of model does	validity	P11, P12

Table 2: The five hypotheses and the pre-specified predictions that operationalise each one. P7 and P4 belong to no hypothesis and are reported in section 4.1 as instrument-validity checks; P4 was overturned on test and its verdict is unchanged in Table C.1. Verdicts are in chapter 4, beside the numbers that decide them.

3.3 Data

FaithBench comprises 750 summaries from ten language models across eight families, annotated at span level under a four-level scheme, with 575 carrying at least one annotation. Under the authors' canonical binarisation, in which Benign counts as faithful, 239 summaries pass and 511 fail. Aggregating the span annotations into one label per summary agrees with the authors' shipped worst-label column on 713 of 717 joinable rows, or 99.44 percent, and the four disagreements are listed in the repository. Using a single dataset is deliberate. RAGTruth (Niu et al., 2024), SummEval (Fabbri et al., 2021) and TRIVIA+ (Chen et al., 2026) were examined and rejected, the last because its contexts run roughly

forty times longer than any locally executable judge can hold, which would have removed the tier the study exists to compare.

3.4 The three deployment tiers

The three arms are ordered by decreasing data exposure, which is the axis an organisation actually chooses along, rather than by any capability ranking. Table 3 lists them with the two properties a procurement decision turns on, whether data leaves the organisation and what the run costs.

Tier	Model	Data leaves org	Weights public	Cost
1 Paid proprietary	claude-opus-5	yes	no	22.21 USD
2 Hosted open weights	gpt-oss:120b-cloud	yes	yes	free tier
3 Local open weights	gpt-oss:20b	no	yes	free

Table 3: The three arms. Cost is what was actually billed for the full run. One auxiliary judge, gemini-3.5-flash-lite, ran on the development split only; it is reported in the appendix and never pooled with the three arms.

The step from tier 2 to tier 3 is controlled: same family, same generation, scale varied alone at roughly 120 billion parameters against 20 billion. That contrast distinguishes open weights judge poorly from what fits on a laptop judges poorly, and only the second is a statement about local deployment. The step from tier 1 to tier 2 is deliberately not controlled. Matching the vendor was considered and rejected, because Nasser (2026) finds same-family judges remain highly distinguishable from their scoring behaviour, so a vendor-matched tier 1 would read as controlled while controlling nothing that varies.

3.5 The judge prompt

The judge returns a binary verdict, faithful or unfaithful, with the span it objects to and a one-sentence justification. It is deliberately not asked for FaithBench's four levels: Questionable encodes annotator uncertainty rather than a property of the text, Benign against Unwanted is a harm judgement layered on a faithfulness judgement, sensitivity and specificity are undefined for a multi-class task without nominating a positive class, and a production gate is binary, so the judge should do the job the gate does. Reasoning is enabled on all three arms and the modes are deliberately not equalised, since the study compares deployments rather than architectures and the vendor's default is part of what a tier delivers. The full prompt is in Appendix A.

3.6 Protocol and the three freezes

All 750 items were judged three times by each judge, giving 6,750 judgments plus a 262-judgment auxiliary run. Rating is absolute rather than pairwise, so position effects do not arise, and temperature is set to zero on the two open-weight tiers and left at the provider default on tier 1, which stopped accepting the parameter, so the three repeats measure the nondeterminism each deployment actually delivers rather than a sampling setting this study imposed on all of them. The corpus splits into 262 development and 488 test items, stratified by summarising model and reference label; stratifying by summariser alone produced a twelve-point difference in the consistent rate between halves.

Three things were fixed before the test split was read, each closing a different route by which the result could have been steered. The prompt was iterated on the development split alone and then frozen at version v0.2 with hash 3b14635aa1c3eecd, which is stored on every judgment row, so the claim that all 7,012 judgments answered the same question can be checked with one query. The analysis code was frozen with file hashes, which limits the freedom to keep trying statistics after seeing the data. And the interpretation was fixed as fourteen numbered predictions, each with a falsifying value. The hashes, the version-control history and the complete predictions are in Appendices A to C. The procedure reduces analytic flexibility; it is not external pre-registration, and section 5.5 says why.

One prompt iteration, version v0.3, was tested on the pilot and rejected because its only measurable effect was to give the weakest judge a hint that let it match the other two. Appendix F records the episode.

3.7 The harness

The harness comprises a loader, the rubric module, three judge adapters behind one interface, a cached and resumable runner, and a SQLite store with one row per judgment. The cache key is the tuple of item, judge, model version and prompt hash, which makes the run resumable without making it unsafe, since a judgment made under a different version or prompt cannot be silently reused. The main run was interrupted twice, by API read timeouts and by a session quota, and neither cost a judgment. The served version string is recorded on every row and compared against the requested one. Figure 1 shows the design: reference labels and judge verdicts are produced along two independent paths that meet only in the analysis.

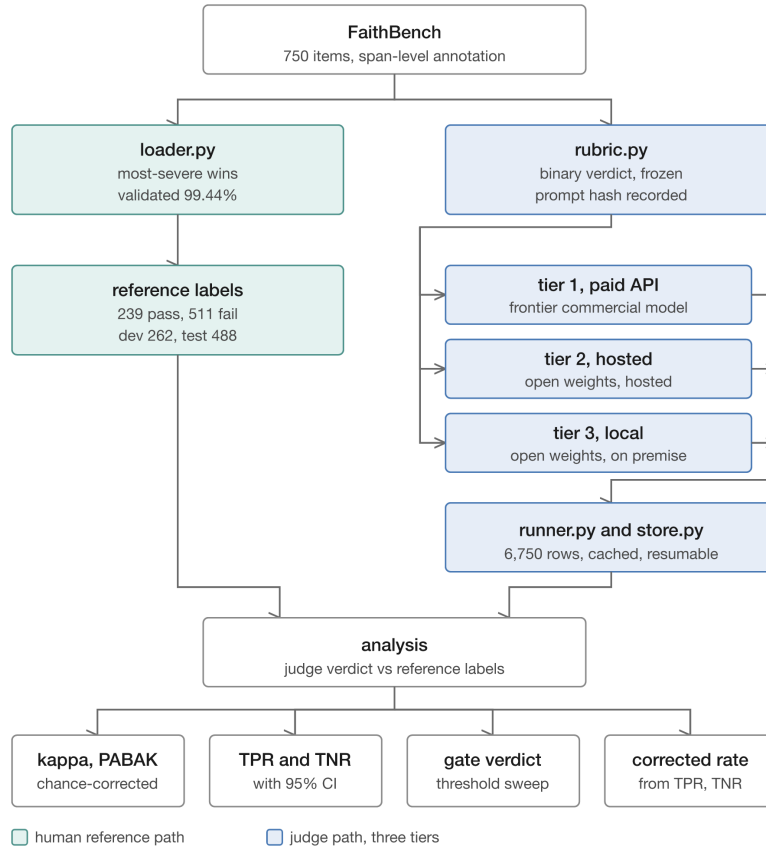


Figure 1: The measurement pipeline. The human reference path aggregates FaithBench’s span annotations to one label per summary using the most-severe-wins rule, then splits the corpus. The judge path sends one frozen rubric, identical across tiers, to three judges chosen to span deployment models. The two meet only in the analysis.

4 Results

All results come from the test split of 488 items, read once on 9 August 2026, after the three freezes described in section 3.6. Table C.1 reproduces the fourteen predictions with their verdicts: ten held, two in part, one weakened to its margin and one was overturned. A prediction is named below only where its verdict changes what can be claimed.

Three statistics carry the chapter. Raw agreement is the proportion of items on which judge and human reference label coincide. Cohen’s kappa (1960) rescales that proportion against the agreement expected from the two raters’ marginal label distributions, so a judge whose verdicts are independent of the reference scores zero however often it happens to match. PABAK (Byrt et al., 1993) is the prevalence-adjusted variant; on a binary task it equals twice raw agreement minus one, which makes it a rescaling of the observed agreement rather than an independent second estimate. The difference between raw agreement and kappa, which the frozen predictions call deflation, is reported descriptively: it indicates

how far the apparent result depends on the statistic chosen, and it is not a count of the agreements chance produced.

4.1 Run quality and reliability of the instrument

The store holds 7,012 judgments, 4,392 of them on the test split. All carry prompt version v0.2 and hash 3b14635aa1c3eecd, and the store holds exactly one distinct pair of those values, so every judgment answered the same question. Requested and served model strings match on every row, so no version drift occurred (Table D.1).

A response that cannot be parsed is a fact about operational reliability rather than about judgment, so the parse-failure rate comes before any agreement statistic. Of 1,464 test judgments each, tier 1 returned 8 unparseable responses, tier 2 returned 43 and tier 3 returned 36, being 0.5, 2.9 and 2.5 percent, and those rows are excluded from every agreement statistic below. The paid tier was therefore about five times more reliable at returning well-formed output, a difference that appears in no agreement statistic in this chapter. Prediction 7 expected failures to rise from tier 1 to tier 3 and they did not, so parse reliability did not track model size.

The failures are also not independent of the reference label: 86 percent of tier 2's unreadable responses fall on items the reference labels mark unfaithful, against a base rate of 68.6 percent, and tiers 1 and 3 sit at 75 percent. Excluding those rows is innocent only if the exclusion is unrelated to the label, so every statistic was recomputed after the freeze under two alternative treatments of them: kappa moves by at most 0.020 and the ordering of the tiers is unchanged (Table G.1).

Repeated runs at a fixed configuration were not deterministic. Kappa across the three repeats spans 0.014, 0.020 and 0.017, against a largest gap between tiers of 0.033; prediction 4 had required the reverse and was overturned. Aggregate stability also conceals instability at the level a gate decides on: all three repeats returned the same verdict on 97.7 percent of tier 1 items, 94.0 percent of tier 3 items and only 84.2 percent of tier 2 items, so roughly one tier 2 item in six received a two-to-one split at an explicit temperature of zero. Those flips largely cancel in the aggregate, which is why the spread of kappa looks small.

4.2 Agreement with the human reference labels

Table 4 reports the primary result under scheme U+Q, the binarisation the frozen rubric gives the judge. Confidence intervals are percentile bootstrap over 2,000 resamples taken over items rather than over judgments, because the three repeats of one item are not independent.

Judge	n	Raw	Kappa	95% CI	PABAK	Diff.	TPR	TNR
Tier 1 paid	1,456	0.466	0.129	[.084, .175]	-0.069	0.337	0.253	0.930
Tier 2 hosted	1,421	0.514	0.162	[.109, .214]	0.029	0.352	0.349	0.868
Tier 3 local	1,428	0.473	0.131	[.083, .178]	-0.055	0.341	0.269	0.916

Table 4: Primary agreement results, scheme U+Q, test split, unparseable judgments excluded. Diff. is raw agreement minus kappa. TPR is sensitivity, the share of reference-unfaithful summaries flagged; TNR is specificity, the share of reference-faithful summaries passed.

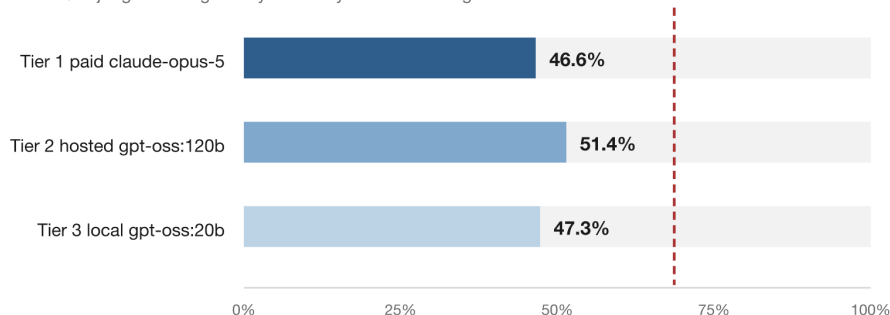
Raw agreement is 46.6 to 51.4 percent and Cohen's kappa is 0.129 to 0.162, a difference of 33.7 to 35.2 percentage points in the same direction on all three tiers. PABAK is -0.069, 0.029 and -0.055. Predictions 1, 2 and 5 had fixed those three ranges in advance and all held.

Table 4 pools the three repeats of each item. Because a gate decides one item at a time, the same statistics were recomputed after the freeze with the repeats aggregated to one verdict per item by majority: raw agreement is then 46.3, 52.2 and 47.7 percent and kappa 0.128, 0.179 and 0.139. Appendix G adds the more conservative rule of flagging an item whenever any repeat flagged it. No conclusion below depends on the aggregation.

Two reference points make the raw figures readable. The first is the composition of the corpus. On the test split 68.6 percent of summaries carry an unfaithful reference label, so a rule that flagged every summary without reading it would reach 68.6 percent raw agreement and a kappa of exactly zero. Figure 2 places the three tiers against that rule, and all three scored below it, which was prediction 13. The comparison is diagnostic rather than competitive: the majority-class rule has zero specificity and no deployment value, and it is used here only to show what a raw agreement figure can conceal.

Raw agreement against the no-skill baseline

test split, scheme U+Q. A judge that flags every summary without reading scores 68.6%. **no skill 68.6%**



All three judges agree with the human labels less often than a rule that reads nothing. Cohen's kappa for that rule is exactly 0.

Figure 2: Raw agreement against the no-skill baseline, which is the majority rule of this corpus. Section 5.4 discusses why the value of that baseline is a property of the corpus and does not transfer to a production stream.

The second reference point answers SQ2. The human annotators agree with one another on 82.0 percent of the binary decisions, measured over 824 annotator pairs on the 443 items carrying redundant annotation, and on 68.0 percent of the four-level decisions. The judges perform the binary task, so 82.0 percent is the comparable figure and all three sit far below it. That comparison is asymmetric, because each judge is scored against a reference aggregated from several annotators while the human figure is one annotator against one other. Table 5 removes the asymmetry by holding out each annotator in turn and scoring them against a reference built from the rest, which is the procedure applied to the judges.

Rater	n	Raw	Kappa	TPR	TNR
Human, leave-one-annotator-out	713	0.823	0.314	0.877	0.474
Tier 1 paid	883	0.327	0.045	0.277	0.955
Tier 2 hosted	850	0.421	0.030	0.396	0.734
Tier 3 local	859	0.342	0.012	0.309	0.750

Table 5: Human annotators and judges scored by an identical procedure on the same 296 test items, those carrying two or more annotators. Post-freeze analysis.

The human figure barely moves under the change of procedure, from 82.0 to 82.3 percent. On these items a held-out annotator reaches a kappa of 0.314 and detects 87.7 percent of the summaries the remaining annotators marked unfaithful, where the judges reach 0.012 to 0.045 and detect 27.7 to 39.6 percent. The subset excludes every summary carrying no annotation, which under the reference rule is exactly the Consistent ones, so it is harder for

both raters and the judge figures are lower than in Table 4. The table supports the comparison between the rows rather than the absolute level of either.

4.3 Error profile and comparison of the deployment tiers

The errors are asymmetric. Sensitivity runs from 0.253 to 0.349 and specificity from 0.868 to 0.930, so the three configurations are conservative rather than indiscriminate: they rarely raise an alarm on a summary the reference labels call faithful, and they miss most of the summaries it calls unfaithful. A single accuracy figure would average the two and report neither. Prediction 6 held. Table 6 splits detection by human severity level.

Judge	Consistent	Benign	Questionable	Unwanted
Tier 1 paid	9%	2%	4%	29%
Tier 2 hosted	12%	18%	8%	39%
Tier 3 local	6%	16%	4%	31%

Table 6: Percentage of judgments at each human severity level that the judge flagged as unfaithful. Under scheme U+Q a low number is correct in the first two columns and a high number in the last two. Items per level: Consistent 113, Benign 40, Questionable 45, Unwanted 290.

Detection on Unwanted items, which FaithBench treats as its most severe category and which formed the largest group in this split at 290 items, reaches 29 to 39 percent, which was prediction 9. Detection was therefore limited on the category the benchmark itself treats as least doubtful, so the shortfall is not accounted for by uncertainty in the Questionable band alone. On that band all three tiers flag less often than they flag Consistent items, at 4 percent against 9 for tier 1, 8 against 12 for tier 2 and 4 against 6 for tier 3, which prediction 8 had named for tier 1. That ordering is reported as observed rather than argued from, since the band holds 45 items and FaithBench defines Questionable as a span that is not clearly a hallucination. The clearer tier difference is on Benign items, where the two open-weight tiers flag 18 and 16 percent against tier 1's 2 percent, raising alarms on content the annotators deliberately passed.

Splitting the same judgments by how far the annotators agreed shows where the agreement in Table 4 comes from: 91.4, 88.4 and 94.3 percent on the 113 items carrying no annotation at all, and between 26.1 and 50.4 percent wherever a human made a judgment call.

All three bootstrap intervals in Table 4 overlap across the band 0.109 to 0.175, which prediction 3 anticipated. Overlapping intervals do not establish that the difference between

two quantities is indistinguishable from zero, so the difference was estimated directly after the freeze, recomputing kappa for each pair of tiers on the same resampled items.

Pair	Difference in kappa	95% CI of the difference
Tier 1 paid minus tier 2 hosted	-0.034	[-0.078, +0.007]
Tier 1 paid minus tier 3 local	+0.002	[-0.044, +0.045]
Tier 2 hosted minus tier 3 local	+0.036	[-0.003, +0.074]

Table 7: Paired difference in Cohen's kappa, 2,000 bootstrap resamples over the 481 test items judged by all three tiers. Post-freeze analysis.

Every interval contains zero, so no pairwise difference in chance-corrected agreement was detected, and none of the intervals is narrow enough to support the stronger claim. Taking 0.03 kappa points as the smallest difference that would matter in practice, no pair falls entirely inside that margin and the interval for tier 2 against tier 3 reaches 0.074. This study therefore did not detect a difference between the tested deployment tiers and is not powered to rule one out. H2 was pre-specified as an expectation of no detectable tier difference, but the design fixed no equivalence margin and used no equivalence-testing procedure, so its result can only be read as no detected difference and not as evidence that the tiers are interchangeable.

One further comparison belongs beside that result. Regrouping the same judgments under FaithBench's other two binarisation schemes moves kappa by up to 0.069, while the largest spread across tiers within a single scheme is 0.065 (Table G.2), so the analyst's choice of binarisation moved the result about as far as the choice of tier did. Prediction 12 had claimed the scheme effect was the larger of the two; the development margin of 0.042 fell to 0.004 on test, so only the weaker statement survives.

4.4 Batch-level release-gate simulation

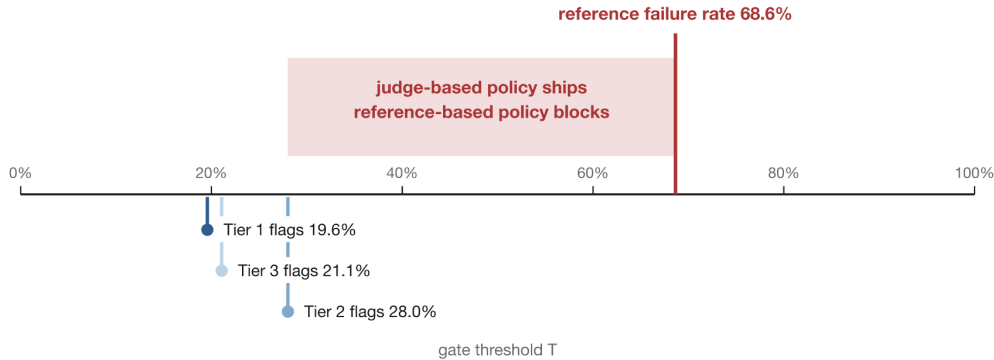
The policy simulated here is the batch-level one defined in section 1.4: a batch of summaries is judged, the share flagged unfaithful is computed, and the release is blocked when that share exceeds a threshold T . The same rule applied to the reference labels blocks whenever T falls below the share they mark unfaithful, which on the test split is 68.6 percent. The two rules are compared at every value of T .

The three tiers flag 19.6, 28.0 and 21.1 percent of test items. Below the lowest flag rate both rules block and above 68.6 percent both rules ship, so the policies agree at the ex-

tremes and disagree in between. For any threshold between 28.0 and 68.6 percent all three tiers recommend shipping while the reference rule recommends blocking, an interval 40.6 percentage points wide against 38.1 points on development, where prediction 10 had required more than 25.

Batch-level release-gate simulation

test split, scheme U+Q. The release is blocked when the flagged share of the batch exceeds the threshold T .



For any T between 28.0% and 68.6%, an interval 40.6 points wide, all three judge-based policies recommend shipping while the reference-based policy recommends blocking.

Figure 3: The batch-level release-gate simulation. Over the shaded interval the judge-based policy recommends shipping while the reference-based policy recommends blocking.

Inside that interval, moving the threshold does not bring the two decisions together. The flag rates show why: every tier flags far fewer items than the reference labels mark unfaithful, so what separates the two policies is how much the judges detect rather than where the threshold sits.

The five hypotheses of Table 2 are now decided. H1 holds, at a difference of 33.7 to 35.2 points. H2 is not contradicted, since no tier difference was detected, but for the reason given in section 4.3 it does not establish equivalence. H3 holds, with specificity above sensitivity on every tier and detection lowest on the ambiguous severity band. H4 holds, at an interval 40.6 points wide. H5 survives only in the weaker form, because the binarisation effect and the tier effect are of the same order rather than the first exceeding the second. The next chapter interprets these results within the limits of the benchmark, the reference labels and the evaluated batch-level policy.

5 Discussion

The findings are interpreted here in the order chapter 4 reported them. Three features of the design bound every reading: the corpus was selected for difficulty, the comparator is ag-

gregated human annotation rather than ground truth, and three deployment configurations were tested rather than a representative sample of LLM judges.

5.1 What the agreement numbers show

The three tested configurations matched the human reference labels on roughly half of the difficult FaithBench items, at 46.6 to 51.4 percent. Once the agreement expected from the two raters' marginal label distributions is accounted for, what remains is a Cohen's kappa of 0.129 to 0.162. The difference of 33.7 to 35.2 points is of similar size on all three tiers.

That difference should be read descriptively. It does not mean that 34 points of individual judgments were caused by chance. Kappa is a chance-corrected coefficient computed from marginal label frequencies, and the gap measures how strongly the apparent result depends on which statistic is reported. Described by raw agreement these judges look mediocre; described by kappa they look close to uninformative. Both statistics describe the same 4,305 usable judgments.

PABAK does not provide independent confirmation of that reading, and none is claimed. Because PABAK on a binary task equals twice raw agreement minus one, its values of -0.069, 0.029 and -0.055 follow arithmetically from raw agreement sitting close to 50 percent. What the pair establishes is narrower and still useful: a prevalence-standardised coefficient also sits near zero, so the low kappa is not solely an artifact of the roughly 69 to 31 class imbalance described by Feinstein and Cicchetti (1990). It does not remove the broader interpretive limits of chance-corrected agreement.

One external comparison is available. Norman, Rivera and Hughes (2026) report a difference of 33 to 41 points between exact-match agreement and Cohen's kappa across roughly 541,000 judgments on MT-Bench. The value measured here is consistent with theirs in direction and magnitude on a different task, corpus and judge set, which extends the concern from general preference judgments to source-summary faithfulness. That study is an unreviewed preprint and this one was not designed as a replication of it, so consistency rather than confirmation is the appropriate claim.

5.2 What the error profile means for a gate

The deployment problem in these numbers is not over-rejection but under-detection. Specificity of 0.868 to 0.930 means all three configurations passed most of the summaries the

reference labels call faithful, while sensitivity of 0.253 to 0.349 means all three missed most of the summaries it calls unfaithful. Tier 1 flagged roughly one reference-unfaithful summary in four, and the highest observed value, 0.349 on tier 2, still leaves about two in three unflagged. A single accuracy figure would average the two into a number that looks tolerable.

The shortfall is not confined to the ambiguous cases. On Unwanted items, the category FaithBench treats as most severe and the one holding 290 of the 488 items, detection reached 29 to 39 percent. The low flag rate on Questionable items is harder to read as straightforward error, since that category is defined to capture annotator uncertainty and rests on 45 items, but setting it aside entirely leaves detection on the unambiguous category still low.

The batch-level simulation turns those rates into a decision. Every tier's flag rate is far below the reference failure rate of 68.6 percent, so for thresholds between 28.0 and 68.6 percent the judge-based policy recommends shipping while the reference-based policy recommends blocking. Moving the threshold inside that interval does not close the gap, because what separates the two policies is how much the judges detect and not where the threshold sits. Retuning is the natural first response to a gate that behaves badly, and on these results it is not the available fix.

The scope of that conclusion belongs with it. It concerns one policy: a binary flag per item, aggregated to a batch rate, compared against a fixed threshold, with no human in the loop. It does not establish that an LLM-assisted quality workflow cannot work on this task, since routing uncertain items to review, replacing the binary verdict with a calibrated score, supplying retrieved evidence and combining several detectors are all untested here. Appendix G indicates that the aggregation rule alone has room in it: flagging an item whenever any of three repeats flagged it raises tier 2 sensitivity from 0.349 to 0.446, at the cost of more false alarms.

5.3 What can and cannot be concluded about the deployment tiers

This study did not detect a difference in chance-corrected agreement between the three tiers, and all three paired intervals in Table 7 contain zero. They are not narrow enough to demonstrate practical equivalence either, since the widest reaches 0.074, outside the 0.03

margin that would count as a difference worth acting on. The appropriate conclusion is not that deployment tier is irrelevant to judge quality, but that 488 items did not resolve a difference between these three configurations.

Agreement was also not the only property that differed, and the properties that did differ are ones a procurement decision can act on. The paid tier returned unparseable output on 0.5 percent of judgments against 2.9 and 2.5 percent for the open-weight tiers, and it flagged 2 percent of Benign items against their 18 and 16 percent. The local tier's median latency was 15.2 seconds against 3.9 seconds for the paid API on the machine documented in Table E.4, and it was the only configuration under which no data left the organisation. A choice between these three cannot rest on kappa alone, and on this evidence kappa is the dimension that separates them least.

5.4 How a difficult-case benchmark shapes the reading

Bao et al. (2025) assembled FaithBench from summaries on which existing automatic detectors disagreed, so every number in chapter 4 is a measurement under difficulty rather than an estimate of performance on a typical production stream. That cuts in both directions.

The first consequence is the 68.6 percent baseline. Because that share of the test split carries an unfaithful reference label, a rule that flags everything without reading reaches 68.6 percent raw agreement and a kappa of exactly zero, and all three tiers scored below it. That rule is a diagnostic majority-class baseline and not a candidate gate, since it blocks every release and has zero specificity. That is a statement about this corpus rather than about summarisation. Where unfaithful outputs are rare the majority rule reverses, flagging nothing becomes the baseline, and a judge has to beat a much higher raw figure while failing on false alarms rather than on misses. What transfers is the test and not the number: a judge should be compared against the majority rule of the corpus it will run on, and against the error costs of the decision it feeds.

The second consequence runs the other way. Difficult cases are where an automated gate has to show that it adds value over simpler checks, so a corpus of difficult cases is a reasonable place to stress one even though it is the wrong place to estimate everyday accura-

cy. These results show that the tested configurations did not hold up under that stress, and they support no claim in either direction about an average production stream.

The third concerns the comparator. What was measured is agreement with aggregated human annotation, not correctness. Ding et al. (2025) examine FaithBench directly and dispute labels on which the annotators had been unanimous, which is a reminder that agreement between annotators establishes consistency rather than truth, and section 2.2 quotes the older version of the same warning from Maynez et al. (2020). The absolute level of agreement in Table 4 is therefore contestable in a way the comparison between the rows of Table 5 is not.

The closest prior work shows how much of a judge's measured quality can come from the scaffolding around it. FaithJudge (Tamber et al., 2025) supplies the judge with nine annotated summaries of the same source document, and with the model held constant balanced accuracy on FaithBench rises from 68.8 to 84.0 percent. That configuration is unavailable to a production gate, which meets each new document without annotated peers of it, so the two sets of numbers answer different questions and the 15.2 point gap indicates what scaffolding is worth here.

5.5 Limitations of interpretation

The reference labels bound the first set of claims. FaithBench provides aggregated human annotation under a four-level scheme in which two levels encode judgment rather than a property of the text, so Questionable and Benign should not be read as equally unambiguous categories, and the binary conversion rule is a choice that moves kappa by up to 0.069. The reference is also imperfect on its own terms, since binary agreement between annotators is 82.0 percent and perfect agreement was never an available target.

The configurations bound the second set. Three deployments are not a sample of LLM judges, and the results depend on the prompt, the model versions, the reasoning modes, the provider defaults, the parsing procedure and the binary output design. Only the step from tier 2 to tier 3 is controlled, holding family and generation fixed while varying scale; tier 1 differs in vendor and architecture as well, for the reason given in section 3.4. Differences therefore cannot be attributed to model size, vendor or hosting location alone, and three conditions are three points rather than a trend.

The simulation bounds the third set, and it qualifies the largest practical claim in the paper. The policy is batch-level rather than item-level, it uses binary flags rather than calibrated scores, and it treats the benchmark reference rate as the comparator. It models no review queue, no review cost and no latency budget, and none of the alternatives listed in section 5.2.

The protocol bounds the fourth. What Appendix C documents is a version-controlled pre-analysis plan rather than a pre-registration in the sense the term carries in clinical research, so the guarantee rests on the commit ordering being checkable rather than on an external witness.

5.6 Implications for evaluation and deployment

The following apply to a practitioner evaluating an LLM judge for a quality-control decision. They follow from the configurations tested here on a corpus selected for difficulty, and are recommendations rather than general findings.

Report class-specific performance rather than one accuracy figure. A judge at 0.93 specificity and 0.25 sensitivity looks acceptable in aggregate while missing three quarters of what a gate exists to catch.

Report a chance-corrected coefficient beside the raw figure. The two differ by roughly 34 points here. Name the binarisation rule as well, since regrouping the labels moved kappa as far as changing the tier did.

Compare the judge against the majority rule of the target corpus. The relevant baseline is a property of the data the judge will see, and a cheap lexical or entailment-based detector is a second useful comparator.

Decide what the judge will trigger before choosing the metric. Blocking an item, routing it to review and scoring a batch carry different error costs, and each implies a different measure.

Validate on a production-like sample before trusting the result in either direction. A difficult benchmark is a stress test. It cannot estimate live prevalence, the false-alarm burden a review team would carry, or the cost of a missed failure.

6 Conclusion and Outlook

6.1 The answer to the research question

On the difficult FaithBench cases evaluated here, raw agreement between judge and human reference labels ranged from 46.6 to 51.4 percent and Cohen's kappa from 0.129 to 0.162. Chance-corrected agreement was therefore low, and the difference of 33.7 to 35.2 points is large enough that the choice of statistic changes how the same judgments read. All three tiers sat below the 68.6 percent majority rule of this corpus and far below the 82.0 percent agreement the annotators reach with one another.

Their central operational weakness was low sensitivity: they flagged 25 to 35 percent of the summaries the reference labels mark unfaithful, while correctly passing 87 to 93 percent of the faithful ones. Under the batch-level policy of section 4.4, flag rates of 19.6, 28.0 and 21.1 percent sit far below the reference failure rate of 68.6 percent, and across the resulting 40.6 point interval no threshold aligns the judge-based release recommendation with the reference-based one. No difference between the tiers was detected. The answer to the second half of the research question is therefore no, for these configurations on this corpus under this policy: the remaining judgment does not support an unaided automatic blocking gate.

6.2 What the study contributes

A measurement contribution. Raw agreement and chance-corrected agreement give materially different accounts of the same judgments here, and so does the binarisation rule, which moved kappa about as far as the choice of tier did. The size of the difference is not itself new: 33.7 to 35.2 points against Norman et al.'s 33 to 41 adds confidence rather than novelty.

An empirical contribution. Three realistic deployment configurations showed low sensitivity on difficult faithfulness cases, with no difference in chance-corrected agreement detected and none ruled out. What separated them was parse reliability, false alarms on benign content, latency and data exposure.

A design contribution. The study documents an evaluation protocol that can be adapted in future LLM-judge work: a held-out split, repeated runs, fixed prompt and analysis specifications, pre-specified predictions, and operational reliability measures reported beside

agreement. Applied here, it overturned one of this paper's own claims and weakened another.

6.3 Outlook

Each of the following follows from a limitation in section 5.5. A prevalence study would establish what these configurations do where unfaithful outputs are rare, which changes the baseline and the cost of a false alarm together. An equivalence test with a margin fixed in advance would settle whether the tiers can be treated as interchangeable, which nothing measured here does.

The policies the simulation excluded are measurable on the data already collected or on one further run: a pass, review or block decision, a calibrated score in place of a binary verdict, and an ensemble over the three tiers. So is the scaffolding effect cited rather than measured here, since one tier and the development split would turn FaithJudge's 68.8 to 84.0 percent rise into a measured number.

Under the prompt, the models, the decision rule and the corpus evaluated here, none of the three tiers detected enough to support an unaided automatic release gate, and the statistic this class of system is usually validated with would not have shown it.

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Appendix

The appendices are the reproducibility evidence and are referenced from the text. Appendix A to E are reproduced in full below. Appendix F, the development record, is the set of development-split tables that exist so the pre-analysis plan in Appendix C can be audited; it ships with the accompanying repository.

Appendix A: the frozen judge prompt

Every judgment in the main run was produced with the exact text below, frozen on 8 August 2026.

Field	Value
PROMPT_VERSION	v0.2
rubric_hash()	3b14635aa1c3eecd
Hash algorithm	sha256(SYSTEM + \x00 + USER), first 16 hex characters
Freeze date	08.08.2026
System prompt length	1241 characters

Table A.1: Prompt identity. The hash is stored on every row of the judgment store, so the claim that all 7,012 judgments used one prompt is checkable with a single query rather than asserted.

System prompt, verbatim

You are a careful fact-checker evaluating whether a summary is faithful to its source document.

A summary is UNFAITHFUL if it contains a claim that:

- contradicts the source document, or
- adds substantive information that the source does not support and that a reader would rely on, such as a name, number, date, outcome, quantity or causal claim that does not appear in the source.

A summary is FAITHFUL if every substantive claim is supported by the source. Not every unsupported detail makes a summary unfaithful. Treat a claim as acceptable when it is unsupported but harmless: a paraphrase, a reasonable generalisation, or a detail that does not change what a reader would take away from the summary. Judge the summary as a whole.

Judge only faithfulness to the source. Do not reward or penalise fluency, length, style, coverage, or how interesting the summary is. Do not use outside knowledge to decide whether a claim is true: the question is whether the source supports it, not whether it is correct in the world.

Respond with a single JSON object and nothing else:
{
 "verdict": "FAITHFUL" | "UNFAITHFUL",
 "span": "<the shortest quoted text from the summary that is unsupported, or null if FAITHFUL>",
 "reason": "<one sentence>"
}

User message template, verbatim

SOURCE DOCUMENT:
{source}

SUMMARY TO EVALUATE:

```
{summary}

Return the JSON object now.
```

The two placeholders are substituted with the stripped text of each item. No other content is sent. In particular the human annotator's span and note are not included, since that would leak the reference label into the judge's own input.

Iteration history. v0.1 was discarded before any run: it defined a summary as unfaithful if any claim was unsupported even if it seemed harmless, which is the strict rule, while the reference labels use FaithBench's canonical binarisation in which Benign counts as faithful. A five-item pilot returned all five Benign items as unfaithful, with every stated reason correct. The judge was not wrong; it was answering the question it had been asked. v0.2 aligns the wording with the canonical binarisation and is the frozen version. v0.3 was drafted, tested on all 60 pilot judgments and rejected, for the reason given in section 3.6.

Appendix B: the frozen analysis

Frozen on 9 August 2026, before the test split was read. The rubric freeze guarantees that every judgment answered the same question. This second freeze guarantees something different and equally necessary, namely that the analysis was fixed before the exam data was seen. Without it the split protects nothing.

File	sha256, first 16
analysis/stats.py	8ad8cc2859a8cc4d
analysis/production.py	259c9cd335bdd91e
src/loader.py	0fa064bf415601c3
src/rubric.py	af8654b9fbba0391

Table B.1: Analysis freeze. Verify with `shasum -a 256` over the four files.

Settled before test was read, on the development split alone: Cohen's kappa as the primary statistic with raw agreement reported alongside it so the difference between them is visible, and PABAK as the prevalence-adjusted check following Byrt et al.'s own recommendation; all three FaithBench binarisation schemes with U+Q primary because it is the rule the frozen rubric gives the judge; percentile bootstrap with 2,000 resamples and seed 20260809, resampled over items rather than judgments; parse failures reported as their own rate and then excluded from every agreement statistic; the four severity levels and the four annotator-agreement strata reported separately; the release gate reported as the inter-

val of thresholds over which the judge-based and the reference-based decision disagree rather than at a single chosen threshold.

Added after the freeze and recorded rather than done quietly: analysis/figures.py, hash 95ef1930ce3c6dab, added on 9 August 2026 after the four files above were frozen and still before test was read. It computes nothing. Every value it draws comes from the frozen statistics module rather than being reimplemented, and the one quantity not returned directly by that module, the expected agreement, is recovered by inverting kappa's own definition rather than recomputed from the corpus prevalence. That is what makes a figure structurally incapable of disagreeing with the table beside it.

Deliberately not decided in advance: which findings the paper leads with. That depends on what test says, and pre-committing to a narrative would be the same error as post-hoc statistic shopping, in the opposite direction.

Appendix C: the fourteen pre-specified predictions

Written on 9 August 2026 before the test split was read, and committed as part of commit 3993094, whose message is Freeze everything before the test split is read, and which holds the pre-test state of the file so that anyone can diff against it. The commit that reads the test split, 58fe1f7, is its direct successor, so the ordering is visible in one log. One thing was appended after the test read and nothing else was touched: the verdict column below. No prediction, threshold or claim has been reworded, softened, removed or added.

A prediction that fails is a result rather than an embarrassment. Fourteen confirmations would have said the split found nothing; a mix is what an honest split looks like. One limit of the procedure is stated here rather than in chapter 5. This is a version-controlled pre-analysis plan, not a pre-registration in the sense the term carries in clinical and psychological research, where a plan is deposited with an independent registry whose timestamp no author can alter. A git history is held by its author, so the guarantee rests on the commit ordering being checkable by diff rather than on an external witness.

#	Prediction for test	Verdict
1	Deflation 30 to 38 pp on all three, same direction	held. 33.7 / 35.2 / 34.1
2	Every arm's kappa in [0.10, 0.25]	held. 0.129 / 0.162 / 0.131
3	All three bootstrap intervals still overlap	held. Common overlap [0.109, 0.175]

#	Prediction for test	Verdict
4	At least one arm's repeat spread exceeds the between-tier gap	overturned. Largest spread 0.020 against a 0.033 gap
5	PABAK within 0.15 of zero on every arm	held. -0.069 / 0.029 / -0.055
6	TNR > TPR on every arm, every TPR below 0.50	held. TNR 0.930 / 0.868 / 0.916
7	Parse failures order tier 1 < tier 2 < tier 3, tier 1 below 1.5%	half held. 0.5 / 2.9 / 2.5, ordering breaks
8	Tier 1's Questionable rate at or below its Consistent rate	held. 4% against 9%
9	Detection on Unwanted between 25 and 50% on every arm	held. 29 / 39 / 31%
10	A divergence band exists and exceeds 25 pp	held, and wider. 40.6 pp
11	Kappa falls monotonically $U > U+Q > U+Q+B$ on every arm	held, tier 1's first step flat
12	The scheme swing still exceeds the between-tier spread	weakened to the edge. 0.069 against 0.065
13	Raw agreement below the test-split prevalence on every arm	held. 46.6 / 51.4 / 47.3% against 68.6%
14	Strata order: no annotation highest, disagreed low-	half held. Disagreed lowest for tier 1 only

Table C.1: Ten held, two half held, one weakened to the edge, one overturned. The claims from the development split that each prediction operationalises are in the repository file; only the test-facing form is reproduced here.

Appendix D: model versions, drift and interruptions

A study that compares three judges is worthless if it cannot show that each judge stayed the same judge for the duration, and naming a model in prose is not a claim a reader can check. Every judgment row therefore records both the model string that was requested and the version string the provider served, and the two are compared automatically.

Judge	Requested	Served	Judgments
tier1_paid	claude-opus-5	claude-opus-5	2,250
tier2_hosted_open	gpt-oss:120b-cloud	gpt-oss:120b-cloud	2,250
tier3_local	gpt-oss:20b	gpt-oss:20b	2,250
tier2_free (auxiliary)	gemini-3.5-flash-lite	gemini-3.5-flash-lite	262

Table D.1: Four rows, one per judge, and zero drift across 7,012 judgments. Had a provider swapped a model mid-run, a fifth row would exist and the affected judgments would be identifiable down to the item.

Two drift incidents, both during setup and both before the main run. First, the paid provider stopped accepting the temperature parameter for the selected model. Sending it produced an error rather than being silently ignored, which is the better failure. The adapter now omits the field, and the consequence is stated in chapter 3 rather than buried: tier 1

runs at the provider's default sampling behaviour while tiers 2 and 3 run at an explicit temperature of zero, so temperature 0 throughout would be an overstatement and is not claimed. Second, a model tag on the open-weight host returned HTTP 410 together with a retirement date. That is the failure mode the version column exists for: a study run three months earlier would have used a model a reader cannot obtain today, and only a recorded version string makes that visible.

Two interruptions during the main run, neither costing a judgment. Repeated read timeouts on the paid tier stopped the run at 59.5 percent of that arm, caused by a 120 second default timeout against a reasoning model and adapters with no retry on transport errors. A session usage limit on the hosted tier stopped it at 78.8 percent against a quota lasting roughly two hours, while the launcher's backoff topped out at twenty minutes. Both were fixed and both were resumed at no cost, because the cache key is the tuple of item, judge, model version and prompt hash, so a judgment made under a different model version or a different prompt could not be silently reused. The lesson worth one sentence is neither bug: it is that both runs failed silently, the wrapper exited zero, the logs stopped, and nothing announced that a nine-hour job had stopped after ninety minutes.

Appendix E: configuration and measured operating cost

	Tier 1 paid	Tier 2 hosted	Tier 3 local
Model requested	claude-opus-5	gpt-oss:120b-cloud	gpt-oss:20b
Host	Anthropic API	Ollama Cloud	Ollama, local
Temperature	not accepted	0.0	0.0
Max output tokens	1000	1000	1000
Reasoning	adaptive, effort high	on	on
Request timeout	300 s	120 s	120 s
Transport retries	3, exp. backoff	3	3
Repeats per item	3	3	3

Table E.1: Configuration per tier. Reasoning modes are deliberately not equalised, because the study compares deployments rather than architectures and the vendor's default is part of what a tier delivers.

Judge	Judgments	Median latency	Input tokens	Output tokens	Median output
Tier 1 paid	2,250	3.9 s	2,658,369	408,672	138
Tier 2 hosted	2,250	5.0 s	1,880,928	1,165,008	496
Tier 3 local	2,250	15.2 s	1,880,928	1,109,610	461

Judge	Judgments	Median latency	Input tokens	Output tokens	Median output
Auxiliary	262	0.7 s	204,666	12,084	39

Table E.2: Measured cost over the full 750-item run, three repeats. Two operational findings appear here and in no agreement statistic: the local tier is roughly four times slower per judgment than the paid API, and the paid tier emits roughly a third of the output tokens for the same verdict.

Cost reconciliation. The store's token totals for tier 1 imply 23.51 USD at list price, being 2,658,369 input tokens at 5.00 USD per million and 408,672 output tokens at 25.00 USD per million. The provider console, read after the run, showed 22.21 USD. The residual is 1.30 USD, or 5.5 percent, and it is unexplained. Prompt caching is ruled out rather than offered as the explanation: no adapter in the harness sets a cache control header, so the harness never requests caching. The direction is also odd, because separate probe databases recorded further paid calls that are not in the main store and should push the console figure above the store's implied total rather than below it. The paper quotes 22.21 USD, because that is what was billed.

Judge	Judgments	Unparseable	Rate
Tier 1 paid	1,464	8	0.5%
Tier 2 hosted	1,464	43	2.9%
Tier 3 local	1,464	36	2.5%

Table E.3: Parse-failure rate per tier, test split, referenced from section 4.1. Unparseable rows are excluded from every agreement statistic in the paper.

Corpus and split. FaithBench, 750 summaries from ten summarising models, span-level human annotation under four levels. Split into 262 development and 488 test items, stratified by summarising model and by reference label. Reference label per summary by most-severe-wins over the span annotations, validated against the authors' own shipped label at 99.44 percent agreement on the 717 joinable rows. Bootstrap: 2,000 percentile resamples, seed 20260809, resampled over items.

Local tier machine and runtime. The 15.2 second median latency is not interpretable without the machine it was measured on, and neither is any claim about whether tier 3 is viable for a given team.

Property	Value
Machine	MacBook Pro, Apple M5
Cores	10, being 4 performance and 6 efficiency

Property	Value
Unified memory	24 GB
Operating system	macOS 26.5.2, build 25F84
Runtime	Ollama 0.32.5
Model tag	gpt-oss:20b, digest 17052f91a42e
Parameters	20.9 billion
Quantisation	MXFP4
Context length	131,072 tokens, runtime default, not overridden
Sampling	temperature 0.0, num_predict 1000
Concurrency	one request at a time, no batching

Table E.4: Local tier execution environment. Machine, operating system and runtime version were read from the machine on 12.08.2026, after the run of 08.08.2026 rather than during it, so an intervening update to the operating system or to Ollama would not be visible here. The model digest, the sampling parameters and the single-request execution are taken from the run log and from the harness source, so those four are contemporaneous with the run.

No other substantial workload was running during the local tier's execution. That is an author statement rather than a measurement, since the harness does not record system load, and it is the one element of this table that cannot be checked from the artifacts.

Appendix F: development record

The development split of 262 items produced its own version of every table in chapter 4. Those tables are the record against which the pre-analysis plan in Appendix C can be audited, and they are marked in the repository as a record rather than as results. They are not reproduced here because reproducing them invites them to be read as findings, which is exactly what the split exists to prevent.

Appendix G: post-freeze robustness

Everything in this appendix was computed after the test split was read and after the analysis freeze, so it is exploratory rather than confirmatory. It exists because several criticisms of the primary analysis are answerable from the data already collected. It is reproduced by `analysis/robustness.py`.

Parse failures. The frozen analysis excludes unparseable judgments, which is innocent only if failure is unrelated to the reference label. Failure rates on unfaithful against faithful items are 0.6 against 0.4 percent on tier 1, 2.7 against 2.0 on tier 3, and 3.7 against 1.3 on tier 2, so the association is material on one arm.

Judge	Unparseable dropped	Counted as flagged	Counted as wrong
Tier 1 paid	0.129	0.129	0.125
Tier 2 hosted	0.162	0.172	0.142
Tier 3 local	0.131	0.133	0.113

Table G.1: Cohen's kappa under three treatments of unparseable output. Column 1 is the frozen analysis. Column 2 routes an unreadable answer to manual review, which is what a cautious gate would do. Column 3 counts it as disagreeing with the human label, the worst case. Post-freeze analysis.

Kappa moves by at most 0.020 across the three treatments and the ordering of the arms is unchanged, so the exclusion flatters tier 2 slightly and changes nothing the paper argues from. Detection of unfaithful items moves in the same narrow way, from 0.253 to 0.258 on tier 1 and from 0.349 to 0.373 on tier 2 under the cautious rule.

Unit of analysis. The primary tables pool three repeats per item, which can be read as though 1,456 independent observations existed where there are 488 items judged up to three times. The bootstrap already resamples items rather than judgments for that reason, but the point estimates are pooled. Aggregating to one verdict per item first, by majority of the three repeats, gives kappa of 0.128, 0.179 and 0.139 against the 0.129, 0.162 and 0.131 reported pooled. Under the more conservative rule of flagging an item if any repeat flagged it, kappa is 0.137, 0.182 and 0.151, and detection of unfaithful items rises to 0.272, 0.446 and 0.313 at the cost of a higher false-alarm rate. No conclusion in chapter 4 depends on which of the three aggregations is used.

Binarisation scheme. The primary result uses U+Q, the rule the frozen rubric gives the judge. The other two schemes regroup the same judgments and cost no further judgments. The largest swing across schemes is 0.069 and the largest spread across tiers within one scheme is 0.065, so the two effects are of the same order.

Judge	U	U+Q (primary)	U+Q+B
Tier 1 paid	0.130	0.129	0.077
Tier 2 hosted	0.195	0.162	0.126
Tier 3 local	0.164	0.131	0.109

Table G.2: Cohen's kappa under all three FaithBench binarisation schemes, test split, with the judgments held fixed. U+Q+B is the scheme FaithJudge uses, which makes that column the one numerically comparable to the closest prior work.